

- For the expert, can combine with modafinil or atomoxetine for ADHD
- For the expert, can occasionally combine with atypical antipsychotics in highly treatment-resistant cases of bipolar disorder or ADHD
- For the expert, can combine with antidepressants to boost antidepressant efficacy in highly treatment-resistant cases of depression while carefully monitoring patient

Tests

- Before treatment, assess for presence of cardiac disease (history, family history, physical exam)
- Blood pressure should be monitored regularly
- In children, monitor weight and height
- Periodic complete blood cell and platelet counts may be considered during prolonged therapy (rare leukopenia and/or anemia)

Side Effects

How Drug Causes Side Effects

- Increases in norepinephrine peripherally can cause autonomic side effects, including tremor, tachycardia, hypertension, and cardiac arrhythmias

- Increases in norepinephrine and dopamine centrally can cause CNS side effects such as insomnia, agitation, psychosis, and substance abuse

Notable Side Effects

- * Insomnia, headache, exacerbation of tics, nervousness, irritability, overstimulation, tremor, dizziness
- Anorexia, nausea, abdominal pain, weight loss
- Can temporarily slow normal growth in children (controversial)
- Blurred vision



Life-Threatening or Dangerous Side Effects

- Psychotic episodes, especially with parenteral abuse
- * Rare priapism
- Seizures
- Palpitations, tachycardia, hypertension
- Rare neuroleptic malignant syndrome
- Rare activation of hypomania, mania, or suicidal ideation (controversial)
- Cardiovascular adverse effects, sudden death in patients with preexisting cardiac structural abnormalities

Weight Gain



- Reported but not expected
- Some patients may experience weight loss

Sedation



- Reported but not expected
- Activation much more common than sedation

What to Do About Side Effects

- Wait
- Adjust dose
- Switch to a formulation of d,l-methylphenidate
- Switch to another agent
- For insomnia, avoid dosing in afternoon/evening

Best Augmenting Agents for Side Effects

- Beta blockers for peripheral autonomic side effects
- Dose reduction or switching to another agent may be more effective since most side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 2.5–10 mg twice per day

Dosage Forms

- Immediate-release tablet 2.5 mg, 5 mg, 10 mg
- Extended-release capsule 5 mg, 10 mg, 15 mg, 20 mg, 25 mg, 30 mg, 35 mg, 40 mg

How to Dose

- Immediate-release: for patients who are not taking racemic d,l-methylphenidate, initial 2.5 mg twice per day in 4-hour intervals; may adjust dose in weekly intervals by 2.5–5 mg/day; maximum dose generally 10 mg twice per day
- Immediate-release: for patients currently taking racemic d,l-methylphenidate, initial dose should be half the current dose of racemic d,l-methylphenidate; maximum dose generally 10 mg twice per day
- Extended-release: for children, same titration schedule as immediate-release but dosed once in the morning; maximum dose 30 mg/day
- Extended-release: for adults not taking racemic d,l-methylphenidate, initial 10 mg/day in the morning; may adjust dose in weekly intervals by 10 mg/day; maximum dose generally 40 mg/day

Dosing Tips

* Immediate-release d-methylphenidate has the same onset of action and duration of action as immediate-release racemic d,l-methylphenidate (i.e., 2–4 hours) but at half the dose

- Extended-release d-methylphenidate contains half the dose as immediate-release beads and half as delayed-release beads, so the dose is released in 2 pulses
- Although d-methylphenidate is generally considered to be twice as potent as racemic d,l-methylphenidate, some studies suggest that the d-isomer is actually more than twice as effective as racemic d,l-methylphenidate
- Side effects are generally dose-related
- Off-label uses are dosed the same as for ADHD

* May be possible to dose only during the school week for some ADHD patients

* May be able to give drug holidays over the summer in order to reassess therapeutic utility and effects on growth and to allow catch-up from any growth suppression as well as to assess any other side effects and the need to reinstitute stimulant treatment for the next school term

- However, most studies show that parental height is what determines a patient's final height, and that most children/adolescents taking stimulants reach their expected height, just more slowly than children/adolescents not exposed to stimulants
- Avoid dosing late in the day because of the risk of insomnia

- Taking with food may delay peak actions for 2–3 hours

Overdose

- Vomiting, tremor, coma, convulsion, hyperreflexia, euphoria, confusion, hallucination, tachycardia, flushing, palpitations, sweating, hyperpyrexia, hypertension, arrhythmia, mydriasis, agitation, delirium, headache

Long-Term Use

- Often used long-term for ADHD when ongoing monitoring documents continued efficacy
- Dependence and/or abuse may develop
- Tolerance to therapeutic effects may develop in some patients
- Long-term stimulant use may be associated with growth suppression in children (controversial)
- Periodic monitoring of weight, blood pressure, CBC, platelet counts, and liver function may be prudent

Habit Forming

- High abuse potential, Schedule II drug
- Patients may develop tolerance, psychological dependence

How to Stop

- Taper to avoid withdrawal effects
- Withdrawal following chronic therapeutic use may unmask symptoms of the underlying disorder and may require follow-up and reinstitution of treatment
- Careful supervision is required during withdrawal from abusive use since severe depression may occur

Pharmacokinetics

- *d-threo* -enantiomer of racemic d,l-methylphenidate
- Mean plasma elimination half-life approximately 2.2 hours (same as d,l-methylphenidate)
- Does not inhibit CYP450 enzymes



Drug Interactions

- May affect blood pressure and should be used cautiously with agents used to control blood pressure
- May inhibit metabolism of SSRIs, anticonvulsants (phenobarbital, phenytoin, primidone), TCAs, and coumarin anticoagulants, requiring downward dosage adjustments of these drugs
- Serious adverse effects may occur if combined with clonidine (controversial)
- Use with MAOIs, including within 14 days of MAOI use, is not advised, but this can sometimes be considered by experts who

monitor depressed patients closely when other treatment options for depression fail

- CNS and cardiovascular actions of d-methylphenidate could theoretically be enhanced by combination with agents that block norepinephrine reuptake, such as the TCAs desipramine or protriptyline, venlafaxine, duloxetine, atomoxetine, milnacipran, and reboxetine
- Theoretically, antipsychotics should inhibit the stimulatory effects of d-methylphenidate
- Theoretically, d-methylphenidate could inhibit the antipsychotic actions of antipsychotics
- Theoretically, d-methylphenidate could inhibit the mood-stabilizing actions of atypical antipsychotics in some patients
- Combinations of d-methylphenidate with mood stabilizers (lithium, anticonvulsants, atypical antipsychotics) is generally something for experts only, when monitoring patients closely and when other options fail
- Antacids or acid suppressants could alter the release of extended-release formulation

Other Warnings/Precautions

- Use with caution in patients with any degree of hypertension, hyperthyroidism, or history of drug abuse

- Children who are not growing or gaining weight should stop treatment, at least temporarily
- May worsen motor and phonic tics
- May worsen symptoms of thought disorder and behavioral disturbance in psychotic patients
- Stimulants have a high potential for abuse and must be used with caution in anyone with a current or past history of substance abuse or alcoholism or in emotionally unstable patients
- Administration of stimulants for prolonged periods of time should be avoided whenever possible or done only with close monitoring, as it may lead to marked tolerance and drug dependence, including psychological dependence with varying degrees of abnormal behavior
- Particular attention should be paid to the possibility of subjects obtaining stimulants for nontherapeutic use or distribution to others and the drugs should in general be prescribed sparingly with documentation of appropriate use
- Usual dosing has been associated with sudden death in children with structural cardiac abnormalities
- Not an appropriate first-line treatment for depression or for normal fatigue
- May lower the seizure threshold
- Emergence or worsening of activation and agitation may represent the induction of a bipolar state, especially a mixed dysphoric bipolar

II condition sometimes associated with suicidal ideation, and require the addition of a mood stabilizer and/or discontinuation of d-methylphenidate

Do Not Use

- If patient has extreme anxiety or agitation
- If patient has motor tics or Tourette's syndrome or if there is a family history of Tourette's, unless administered by an expert in cases when the potential benefits for ADHD outweigh the risks of worsening tics
- Should generally not be administered with an MAOI, including within 14 days of MAOI use, except in heroic circumstances and by an expert
- If patient has glaucoma
- If patient has structural cardiac abnormalities
- If patient has angioedema or anaphylaxis
- If there is a proven allergy to methylphenidate

Special Populations

Renal Impairment

- No dose adjustment necessary

Hepatic Impairment

- No dose adjustment necessary

Cardiac Impairment

- Use with caution, particularly in patients with recent myocardial infarction or other conditions that could be negatively affected by increased blood pressure
- Do not use in patients with structural cardiac abnormalities

Elderly

- Some patients may tolerate lower doses better

Children and Adolescents

- Safety and efficacy not established in children under age 6
- Use in young children should be reserved for the expert
- Methylphenidate has acute effects on growth hormone; long-term effects are unknown but weight and height should be monitored during long-term treatment
- Half-life and duration of clinical action tend to be shorter in younger children
- Usual dosing has been associated with sudden death in children with structural cardiac abnormalities
- American Heart Association recommends ECG prior to initiating stimulant treatment in children, although not all experts agree

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
 - Controlled studies have not been conducted in pregnant women
 - Infants whose mothers took methylphenidate during pregnancy may experience withdrawal symptoms
 - Racemic methylphenidate has been shown to have teratogenic effects in rabbits when given in doses of 200 mg/kg/day throughout organogenesis
 - Use in women of childbearing potential requires weighing potential benefits to the mother against potential risks to the fetus
- * For ADHD patients, methylphenidate should generally be discontinued before anticipated pregnancies

Breast Feeding

- Unknown if methylphenidate is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed
- If infant shows signs of irritability, drug may need to be discontinued

The Art of Psychopharmacology

Potential Advantages

- The active d enantiomer of methylphenidate may be slightly more than twice as efficacious as racemic d,l-methylphenidate

Potential Disadvantages

- Patients with current or past substance abuse, bipolar disorder, or psychosis

Primary Target Symptoms

- Concentration, attention span
- Motor hyperactivity
- Impulsiveness
- Physical and mental fatigue
- Daytime sleepiness
- Depression



Pearls

- May be useful for treatment of depressive symptoms in medically ill elderly patients
- May be useful for treatment of post-stroke depression

- A classical augmentation strategy for treatment-refractory depression
- Specifically, may be useful for treatment of cognitive dysfunction and fatigue as residual symptoms of major depressive disorder unresponsive to multiple prior treatments
- May also be useful for the treatment of cognitive impairment, depressive symptoms, and severe fatigue in patients with HIV infection and in cancer patients
- Can be used to potentiate opioid analgesia and reduce sedation, particularly in end-of- life management
- Atypical antipsychotics may be useful in treating stimulant or psychotic consequences of overdose
- Some patients respond to or tolerate methylphenidate better than amphetamine, and vice versa
- Drug abuse may actually be lower in ADHD adolescents treated with stimulants than in ADHD adolescents who are not treated
- New extended-release formulation is truly a once daily dose
- Extended-release capsule can be sprinkled over applesauce for patients unable to swallow the capsule
- Some patients may benefit from an occasional addition of an immediate-release dose of d-methylphenidate to the daily base dose of extended-release d-methylphenidate

Suggested Reading

Dexmethylphenidate –Novartis/Celgene. Focalin, D-MPH, D-methylphenidate hydrochloride, D-methylphenidate, dexmethylphenidate, dexmethylphenidate hydrochloride . *Drugs R D* 2002 ;**3** (4):279–82 .

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Keating GM , Figgitt DP . Dexmethylphenidate . *Drugs* 2002 ;**62** (13):1899–904.

Methylphenidate (D,L)

Therapeutics

Brands

- Concerta
- Metadate CD
- Ritalin
- Ritalin LA
- Methylin
- QuilliChew ER
- Quillivant XR
- Aptensio XR
- Daytrana
- Jornay PM
- Adhansia XR
- Cotempla XR-ODT

see index for additional brand names

Generic?

Yes

Class

- Neuroscience-based Nomenclature: dopamine, norepinephrine reuptake inhibitor and releaser (DN-RIRe)
- Stimulant

Commonly Prescribed for

(bold for FDA approved)

- **Attention deficit hyperactivity disorder (ADHD) in children and adults (approved ages vary based on formulation)**
- **Narcolepsy (Metadate ER, Methylin ER, Ritalin, Ritalin SR)**
- Treatment-resistant depression

How the Drug Works

- * Increases norepinephrine and especially dopamine actions by blocking their reuptake
- Enhancement of dopamine and norepinephrine actions in certain brain regions (e.g., dorsolateral prefrontal cortex) may improve attention, concentration, executive function, and wakefulness
- Enhancement of dopamine actions in other brain regions (e.g., basal ganglia) may improve hyperactivity
- Enhancement of dopamine and norepinephrine in yet other brain regions (e.g., medial prefrontal cortex, hypothalamus) may improve depression, fatigue, and sleepiness

How Long Until It Works

- Some immediate effects can be seen with first dosing
- Can take several weeks to attain maximum therapeutic benefit

If It Works (for ADHD)

- The goal of treatment of ADHD is reduction of symptoms of inattentiveness, motor hyperactivity, and/or impulsiveness that disrupt social, school, and/or occupational functioning
- Continue treatment until all symptoms are under control or improvement is stable and then continue treatment indefinitely as long as improvement persists
- Reevaluate the need for treatment periodically
- Treatment for ADHD begun in childhood may need to be continued into adolescence and adulthood if continued benefit is documented

If It Doesn't Work (for ADHD)

- Consider adjusting dose or switching to another formulation of d,l-methylphenidate or to another agent
- Consider behavioral therapy
- Consider the presence of noncompliance and counsel patient and parents
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., bipolar disorder, substance abuse, medical illness,

etc.)

* Some ADHD patients and some depressed patients may experience lack of consistent efficacy due to activation of latent or underlying bipolar disorder, and require either augmenting with a mood stabilizer or switching to a mood stabilizer



Best Augmenting Combos for Partial Response or Treatment Resistance

- * Best to attempt other monotherapies prior to augmenting
- For the expert, can combine immediate-release formulation with a sustained-release formulation of d,l-methylphenidate for ADHD
- For the expert, can combine with modafinil or atomoxetine for ADHD
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Notable Side Effects

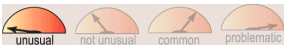
- * Insomnia, headache, exacerbation of tics, nervousness, irritability, overstimulation, tremor, dizziness
- Anorexia, nausea, abdominal pain, weight loss
- Can temporarily slow normal growth in children (controversial)
- Blurred vision
- Transdermal: application site reactions, including contact sensitization (erythema, edema, papules, vesicles) and chemical leukoderma



Life-Threatening or Dangerous Side Effects

- * Rare priapism
- Psychotic episodes, especially with parenteral abuse
- Seizures
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Dosing and Use

Usual Dosage Range

- ADHD (oral): varies by formulation; see How to Dose section
- ADHD (transdermal): 10–30 mg/9 hours
- Narcolepsy: 20–60 mg/day in 2–3 divided doses

Dosage Forms

- Immediate-release tablet 5 mg, 10 mg, 20 mg (Ritalin, generic methylphenidate)
- Oral solution 5 mg/mL, 10 mg/5 mL (Methylin)

- Older sustained-release tablet 10 mg, 20 mg (Methylin ER); 20 mg (Ritalin SR, Metadate ER)
- * Newer sustained-release capsule 10 mg, 20 mg, 30 mg, 40 mg, 60 mg (Ritalin LA); 10 mg, 20 mg, 30 mg, 40 mg, 50 mg, 60 mg (Metadate CD)
- * Newer sustained-release tablet 18 mg, 27 mg, 36 mg, 54 mg (Concerta)
- Sustained-release chewable tablet 20 mg scored, 30 mg, 40 mg (QuilliChew ER)
- Extended-release capsule, multi-layer release 10 mg, 15 mg, 20 mg, 30 mg, 40 mg, 50 mg, 60 mg (Aptensio XR)
- Orally disintegrating tablet 8.6 mg, 17.3 mg, 25.9 mg (Cotempla XR-ODT)
- Extended-release capsule 20 mg, 40 mg, 60 mg, 80 mg, 100 mg (Jornay PM)
- Extended-release capsule 25 mg, 35 mg, 45 mg, 55 mg, 70 mg, 85 mg (Adhansia XR)
- Extended-release oral suspension 5 mg/mL (Quillivant XR)
- Transdermal patch 27 mg/12.5 cm² (10 mg/9 hours; 1.1 mg/hour), 41.3 mg/18.75 cm² (15 mg/9 hours; 1.6 mg/hour), 55 mg/25 cm² (20 mg/9 hours; 2.2 mg/hour), 82.5 mg/37.5 cm² (30 mg/9 hours; 3.3 mg/hour)

How to Dose

- Immediate-release Ritalin, generic methylphenidate (2–4 hour duration of action)
 - ADHD: initial 5 mg in morning, 5 mg at lunch; can increase by 5–10 mg each week; maximum dose generally 60 mg/day
 - Narcolepsy: give each dose 30–45 minutes before meals; maximum dose generally 60 mg/day
- Older extended-release Ritalin SR, Methylin SR, and Metadate ER
 - These formulations have a duration of action of approximately 4–6 hours; therefore, these formulations may be used in place of immediate-release formulations when the 4–6 hour dosage of these sustained-release formulations corresponds to the titrated 4–6 hour dosage of the immediate-release formulation
 - Average dose is 20–30 mg/day, usually in 2 divided doses

* Newer sustained-release formulations for ADHD

- Concerta (up to 12 hours duration of action): initial 18 mg/day in morning; can increase by 18 mg each week; maximum dose generally 72 mg/day
- Ritalin LA, Metadate CD, QuilliChew ER (up to 8 hours duration of action): initial 20 mg once daily; dosage may be adjusted in weekly increments of 10 mg or 20 mg (QuilliChew ER only) to a maximum of 60 mg/day taken in the morning
- Quillivant XR (up to 12-hour duration): initial 20 mg once daily; dosage may be adjusted in weekly increments of 10–20 mg to a maximum of 60 mg/day taken in the morning

- Aptensio XR (up to 12-hour duration): initial 10 mg once daily; dosage may be adjusted in weekly increments of 10 mg to a maximum of 60 mg/day taken in the morning
- Cotelpla XR-ODT (up to 12-hour duration): initial dose 17.3 mg once daily in the morning; can increase weekly in increments of 8.6 mg to 17.3 mg per day; maximum recommended dose 51.8 mg; should be taken consistently either with or without food
- Jornay PM (nighttime dosing): initial dose 20 mg in the evening at 8 pm; can increase in weekly increments of 20 mg per day; maximum daily dose 100 mg; can adjust administration time between 6:30 to 9:30, depending on optimal efficacy and tolerability; once optimal administration time is set it should remain consistent
- Adhansia XR: initial dose 25 mg once in the morning; can increase every 5 days in increments of 10–15 mg per day; maximum daily dose 85 mg in adults and 70 mg in pediatric patients; doses above 85 mg in adults and 70 mg in pediatric patients are associated with increased risk of adverse effects

* Transdermal formulation for ADHD

- Initial 10 mg/9 hours; can increase by 5 mg/9 hours every week; maximum dose generally 30 mg/9 hours
- Patch should be applied 2 hours before effect is needed and should be worn for 9 hours